Module 2

Introduction

- Data mining functionalities are used to specify the kind of patterns to be found in data mining tasks.
- Data mining tasks:
  - **Descriptive data mining**: characterize the general properties of the data in the database.
  - **Predictive data mining**: perform inference on the current data in order to make predictions.

Introduction

- Different views lead to different classifications
  - **Data view**: Kinds of data to be mined
  - **Knowledge view**: Kinds of knowledge to be discovered
  - **Method view**: Kinds of techniques utilized
  - **Application view**: Kinds of applications adapted

Mining Frequent Patterns, Associations, and Correlations

Frequent Patterns

- **Frequent patterns**,
  - are patterns that occur frequently in data.
- The kinds of frequent patterns:
  - frequent itemsets patterns
  - frequent sequential patterns
  - frequent structured patterns

Frequent Patterns

- **A frequent itemset pattern**
  - refers to a set of items that frequently appear together in a transactional data set, such as milk and bread.
- **A frequent sequential pattern**
  - such as the pattern that customers tend to purchase first a PC, followed by a digital camera, and then a memory card, is a (frequent) sequential pattern.
- **A frequent structured pattern**
  - can refer to different structural forms, such as graphs, trees, or lattices, which may be combined with itemsets or subsequences.
  - If a substructure occurs frequently, it is called a (frequent) structured pattern.

Frequent Patterns

- Mining frequent patterns leads to the discovery of interesting **associations** and **correlations** within data.

Association Analysis

- Suppose, as a marketing manager of *AllElectronics*, you would like to determine which items are frequently purchased together within the same transactions.
- An example from the AllElectronics transactional database, is:

$$buys(X, \text{"computer"}) \Rightarrow buys(X, \text{"software"}) \quad [support = 1\%, confidence = 50\%]$$

  - where X is a variable representing a customer.

Association Analysis

- A **confidence**, or **certainty**, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well.
- A **1% support** means that 1% of all of the transactions under analysis showed that computer and software were purchased together.
- This association rule involves **a single attribute** or predicate (i.e., buys) that repeats.
- Association rules that contain a single predicate are referred to as **single-dimensional association rules**.

Association Analysis

- We may find association rules like:

$$age(X, \text{"20...29"}) \wedge income(X, \text{"20K...29K"}) \Rightarrow buys(X, \text{"CD player"})$$

$$[support = 2\%, confidence = 60\%]$$

  - The rule indicates that of the AllElectronics customers under study, 2% are 20 to 29 years of age with an income of 20,000 to 29,000 and have purchased a CD player at *AllElectronic*
  - There is a 60% probability that a customer in this age and income group will purchase a CD player.
- This is an association between more than one attribute (i.e., age, income, and buys).
- This is **a multidimensional association rule**.

Support and confidence of a rule

- Example: 4 cool days with normal humidity

```
If temperature = cool then humidity = normal
```

  - Support = 4, confidence = 100%
- Normally: minimum support and confidence prespecified, e.g. support >= 2 and confidence >= 95% for weather data

Interpreting association rules

- Interpretation is not obvious:

```
If windy = false and play = no then outlook = sunny
                                 and humidity = high
```

- is **not** the same as

```
If windy = false and play = no then outlook = sunny
If windy = false and play = no then humidity = high
```

- It means that the following also holds:

```
If humidity = high and windy = false and play = no
   then outlook = sunny
```

Association Analysis

- Large number of possible associations
  - Output needs to be restricted to show only the most predictive associations
- Association rules are interesting if they do satisfy both:
  - **A minimum support threshold:** number of instances predicted correctly
  - **A minimum confidence threshold:** number of correct predictions, as proportion of all instances that rule applies to

Association Analysis

- Association learning is unsupervised
- Association rules usually involve only nonnumeric attributes
- Difference to classification learning:
  - Can predict any attribute’s value, not just the class
  - More than one attribute’s value at a time
  - There are more association rules than classification rules
- Additional analysis can be performed to uncover interesting **statistical correlations** between associated attribute-value pairs.

Classification

Classification

- **Classification**
  – Construct models (functions) that describe and distinguish classes or concepts to predict the class of objects whose **class label** is unknown
- Example:
  – In weather problem the play or don’t play judgment
  – In contact lenses problem the lens recommendation
- **Training data set**
  – The derived model is based on the analysis of a **set of training data** (i.e., data objects whose class label is known).

Classification

- The success of classification learning
  - Using an independent set of **test data** for which class labels are known but not made available to the machine.
- Classification learning is supervised
  - Process is provided with actual outcome

Classification

- **Examples:**
  - Decision Trees
  - Classification Rules
  - Neural Network
  - Naïve Bayesian classification,
  - Support vector machines
  - k-nearest neighbor classification

Decision trees

- “Divide and conquer” approach produces decision tree
- Nodes involve testing a particular attribute
- Usually, attribute value is compared to constant
- Other possibilities:
  - Comparing values of two attributes
  - Using a function of one or more attributes

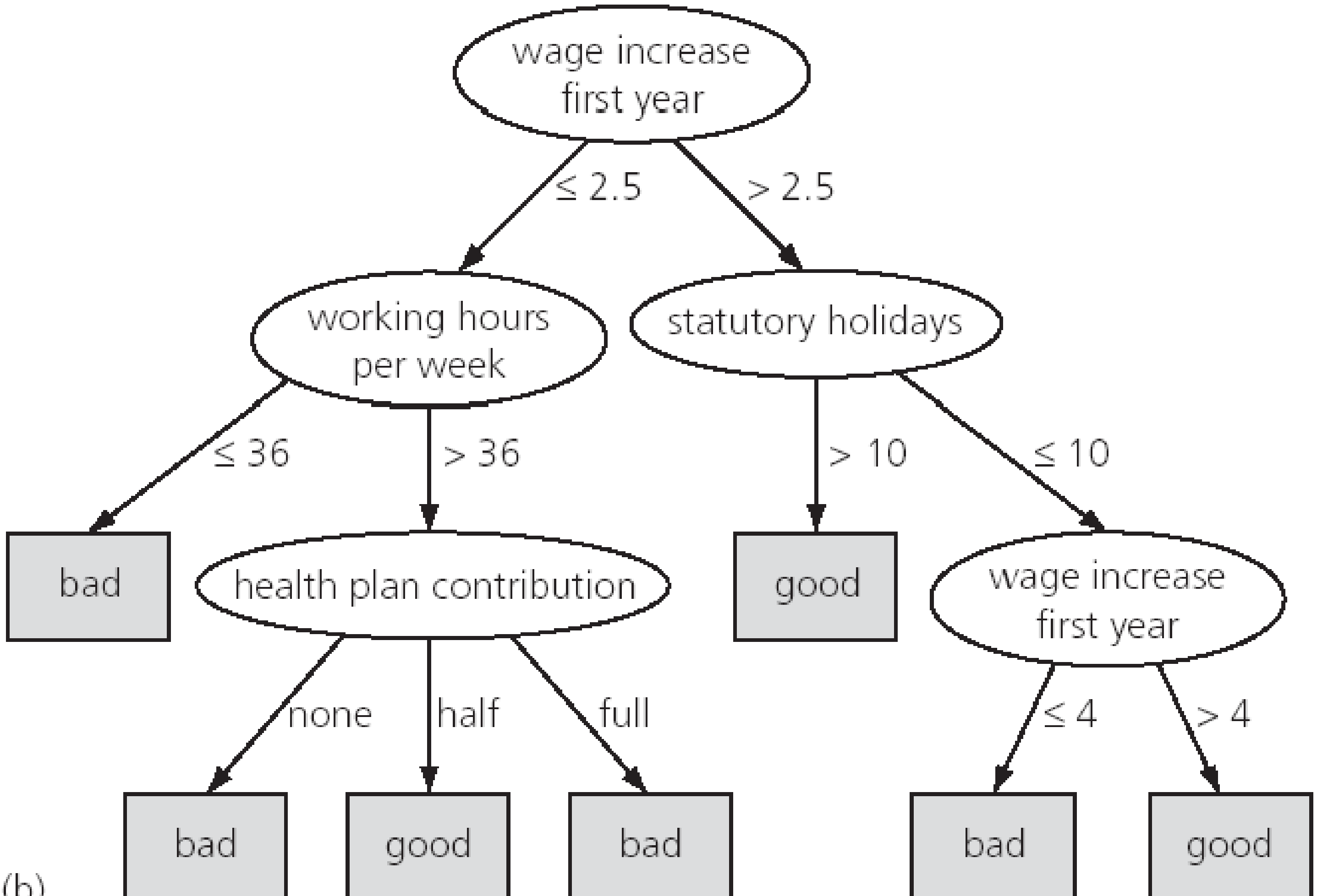
Decision trees

- Leaf nodes
  - give a classification that applies to all instances that reach the leaf
  - set of classifications
  - probability distribution over all possible classifications
- To classify an unknown instance,
  - it is routed down the tree according to the values of the attributes tested in successive nodes, and
  - when a leaf is reached the instance is classified according to the class assigned to the leaf.

Nominal and numeric attributes

- If the attribute is nominal:
  - Number of children usually equal to the number of possible values
  - Usually attribute won't get tested more than once
- If the attribute is numeric:
  - Test whether value is greater or less than constant
  - Attribute may get tested several times
  - Other possibility: three-way split (or multi-way split)
    - Integer: less than, equal to, greater than
    - Real: below, within, above

Decision tree for the labor data



(b)

Classification rules

- Popular alternative to decision trees
- Rules include two parts:
  - **Antecedent or precondition**:
    - a series of tests just like the tests at the nodes of a decision tree
    - Tests are usually logically ANDed together
    - All the tests must succeed if the rule is to fire
  - **Consequent or conclusion**:
    - The class or set of classes or probability distribution assigned by rule
- Example: A rule from contact lens problem

```
If tear production rate = reduced then recommendation = none
```

From trees to rules

- Easy: converting a tree into a set of rules
  - One rule for each leaf:
    - Antecedent contains a condition for every node on the path from the root to the leaf
    - Consequent is class assigned by the leaf
- Produces rules that are very clear
  - Doesn't matter in which order they are executed
- But: resulting rules are unnecessarily complex
  - It needs to remove redundant tests/rules

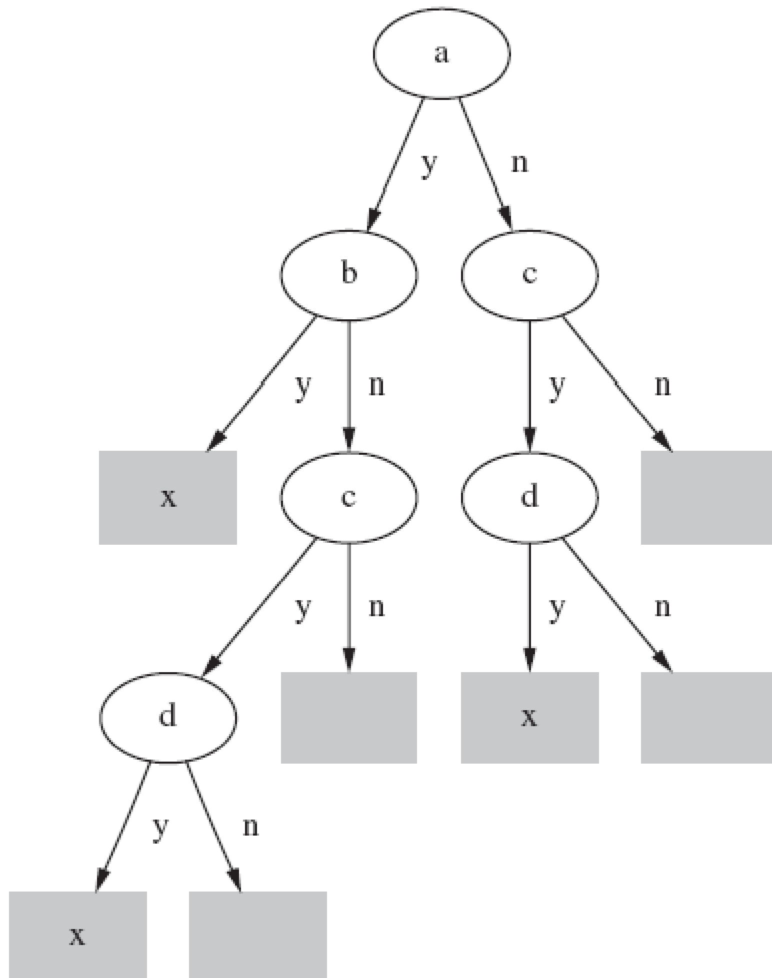
From rules to trees

- More difficult: transforming a rule set into a tree
  - Tree cannot easily express disjunction between rules
- Example: rules which test different attributes

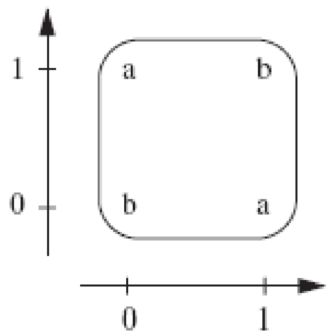
```
If a and b then x
If c and d then x
```

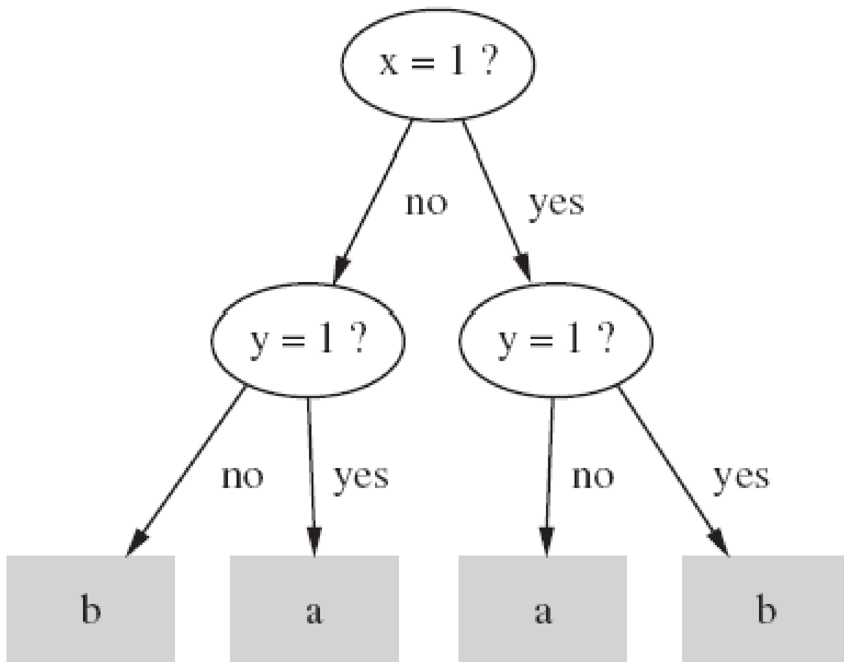
A tree for a simple disjunction

- **Replicated subtree problem:** tree contains identical subtrees



The exclusive-or problem





If x=1 and y=0 then class = a

If x=0 and y=1 then class = a

If x=0 and y=0 then class = b

If x=1 and y=1 then class = b

A tree with a replicated subtree

- If it is possible to have a "default" rule that covers cases not specified by the other rules, rules are much more compact than trees
- There are four attributes, *x, y, z,* and *w,* each can be 1, 2, or 3

If x=1 and y=1 then class = a

If z=1 and w=1 then class = a

Otherwise class = b

Cluster Analysis

Cluster Analysis

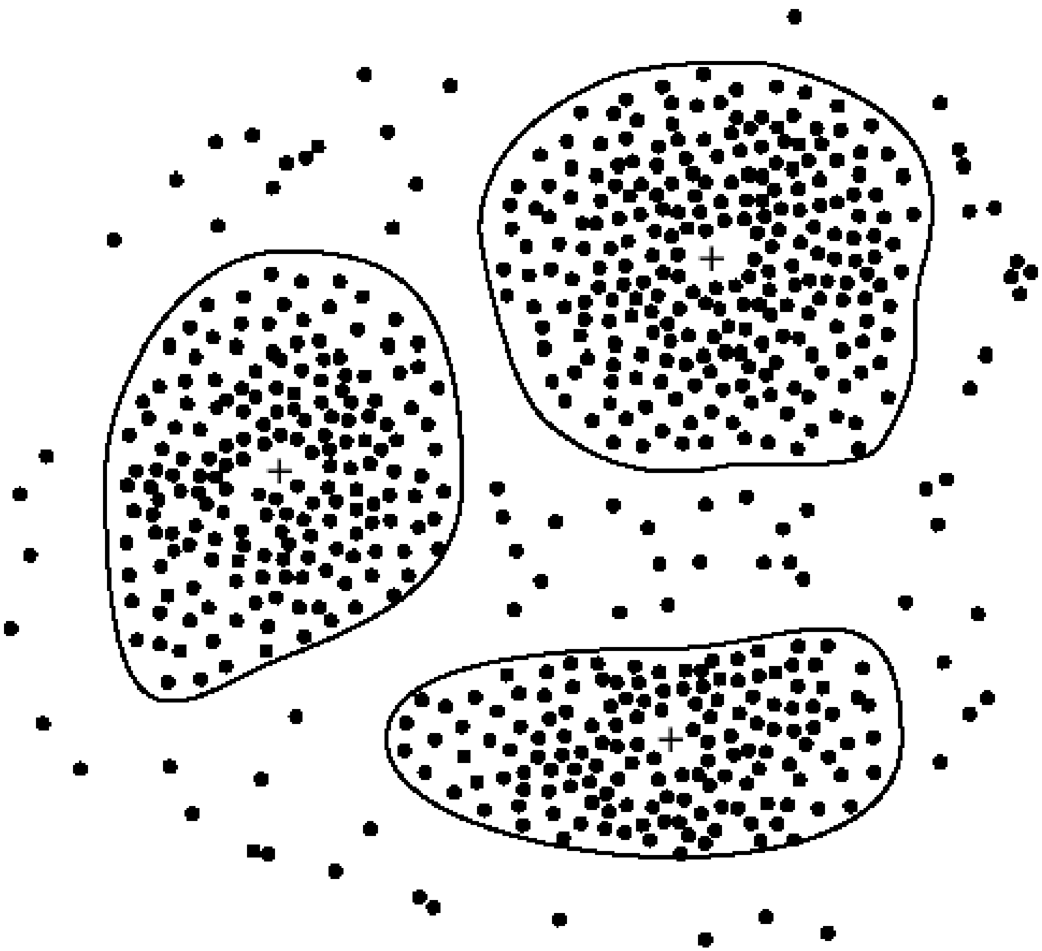
- **Clustering**:
  - grouping similar instances into clusters
- Clustering is unsupervised
  - The class of an example is not known
- Example:
  - a version of the iris data in which the type of iris is omitted
  - Then it is likely that the 150 instances fall into natural clusters corresponding to the three iris types.

Iris data as a clustering problem

	Sepal length (cm)	Sepal width (cm)	Petal length (cm)	Petal width (cm)
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
. . .				
51	7.0	3.2	4.7	1.4
52	6.4	3.2	4.5	1.5
53	6.9	3.1	4.9	1.5
54	5.5	2.3	4.0	1.3
55	6.5	2.8	4.6	1.5
. . .				
101	6.3	3.3	6.0	2.5
102	5.8	2.7	5.1	1.9
103	7.1	3.0	5.9	2.1
104	6.3	2.9	5.6	1.8
105	6.5	3.0	5.8	2.2
. . .				

Cluster Analysis

- A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters. Each cluster "center" is marked with a "+".



Clustering

- Clustering may be followed by a second step of classification learning in which rules are learned that give an intelligible description of how new instances should be placed into the clusters.

Interesting Patterns

Interesting Patterns

- Data mining may generate thousands of patterns: Not all of them are interesting
- **What makes a pattern interesting?**
  1. Easily understood by humans,
  2. Valid on new or test data with some degree of certainty
  3. Potentially useful
  4. Novel
  5. Validates some hypothesis that a user seeks to confirm

Interesting Patterns

- Objective vs. subjective interestingness measures
  - Objective: based on statistics and structures of patterns, e.g., support, confidence, etc.
  - Subjective: based on user's belief in the data, e.g., unexpectedness, novelty, actionability, etc.

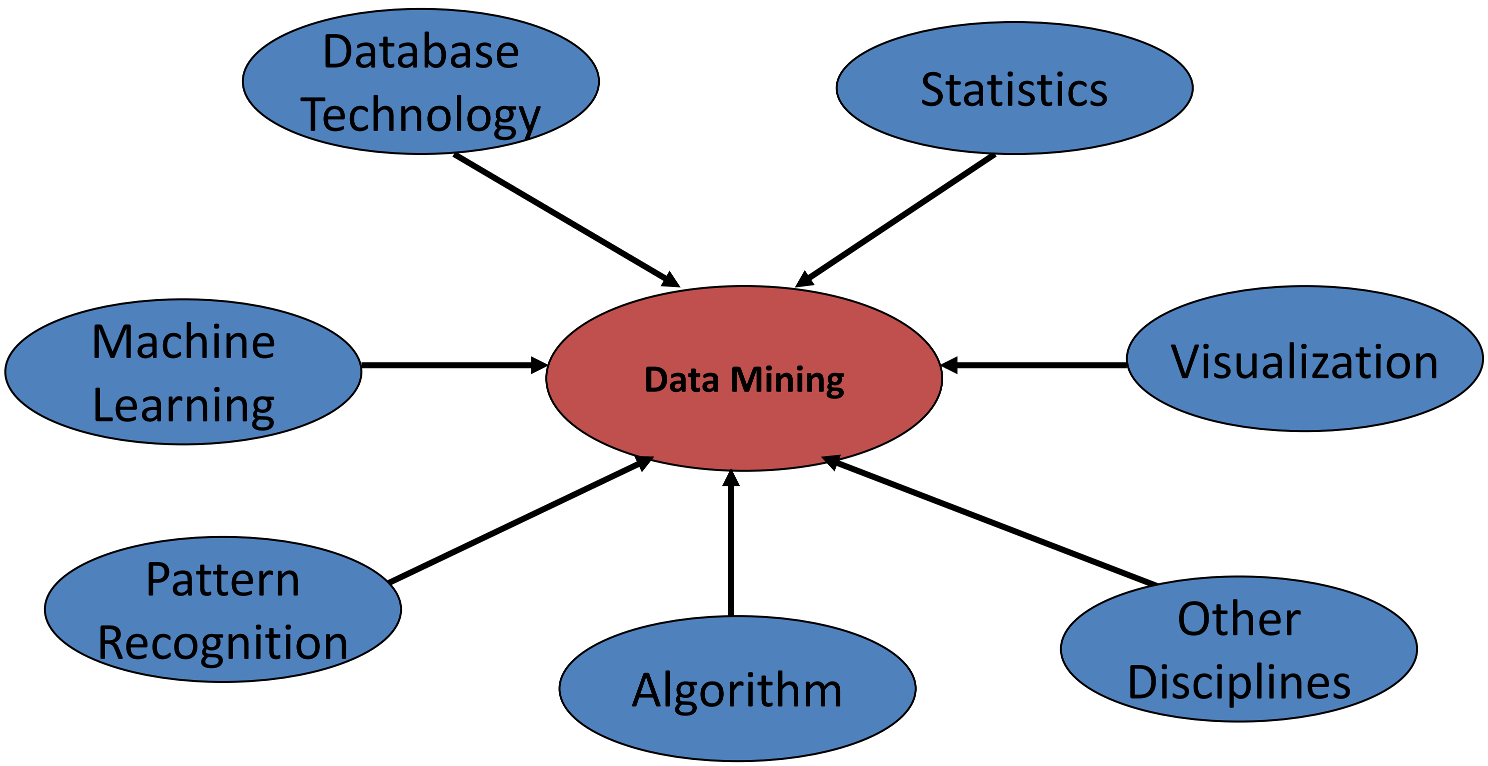
Interesting Patterns

- **Can a data mining system generate all of the interesting patterns?**
  - refers to the **completeness** of a data mining algorithm
  - Do we need to find all of the interesting patterns?
  - **Association rule** mining is an example (where the use of constraints and interestingness measures) can ensure the completeness of mining.
  - Heuristic vs. exhaustive search

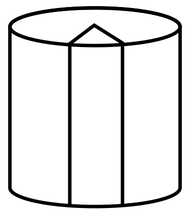
Interesting Patterns

- **Can a data mining system generate only interesting patterns?**
  - First general all the patterns and then filter out the uninteresting ones
  - Generate only the interesting patterns

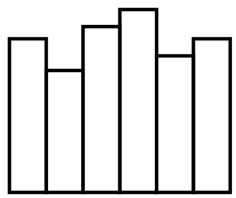
Classification of Data Mining Systems



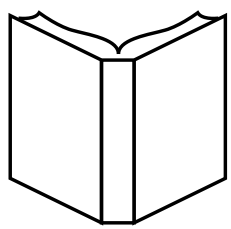
Data Mining Task Primitives



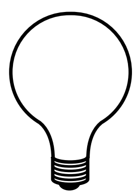
Task-relevant data
Database or data warehouse name
Database tables or data warehouse cubes
Conditions for data selection
Relevant attributes or dimensions
Data grouping criteria



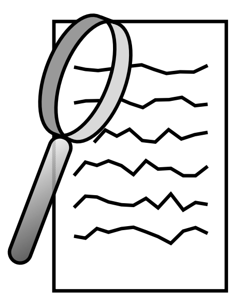
Knowledge type to be mined
Characterization
Discrimination
Association/correlation
Classification/prediction
Clustering



Background knowledge
Concept hierarchies
User beliefs about relationships in the data



Pattern interestingness measures
Simplicity
Certainty (e.g., confidence)
Utility (e.g., support)
Novelty



Visualization of discovered patterns
Rules, tables, reports, charts, graphs, decision trees, and cubes
Drill-down and roll-up

Integration Schemes

- **No Coupling** − In this scheme, the data mining system does not utilize any of the database or data warehouse functions. It fetches the data from a particular source and processes that data using some data mining algorithms. The data mining result is stored in another file.
- **Loose Coupling** − In this scheme, the data mining system may use some of the functions of database and data warehouse system. It fetches the data from the data respiratory managed by these systems and performs data mining on that data. It then stores the mining result either in a file or in a designated place in a database or in a data warehouse.
- **Semi−tight Coupling** − In this scheme, the data mining system is linked with a database or a data warehouse system and in addition to that, efficient implementations of a few data mining primitives can be provided in the database.
- **Tight coupling** − In this coupling scheme, the data mining system is smoothly integrated into the database or data warehouse system. The data mining subsystem is treated as one functional component of an information system.

Major Issues in Data Mining

Major issues in data mining research, partitioning them into five groups:

1) *Mining methodology*
2) *User interaction*
3) *Efficiency and scalability*
4) *Diversity of data types*
5) *Data mining and society*